



BITS Pilani
K K Birla Goa Campus

8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

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Address Decoding

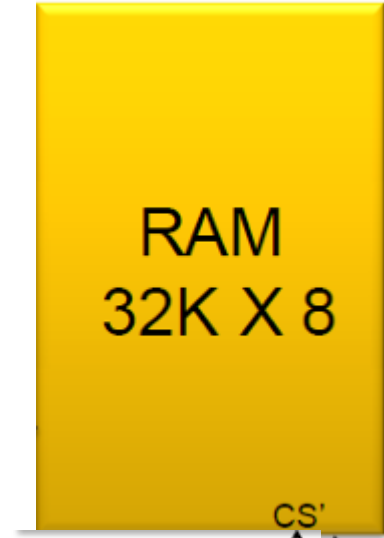
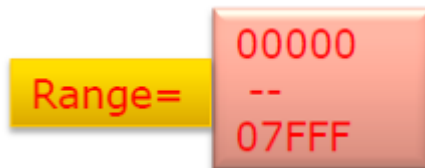
Address decoding can be done using:

- Logic Gates
- Decoder (2 to 4, 3 to 8)
- Programmable logic (PLAs, PLDs, FPGAs)

Address decoding

Design an address decoder for a 32K x 8 RAM.

A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

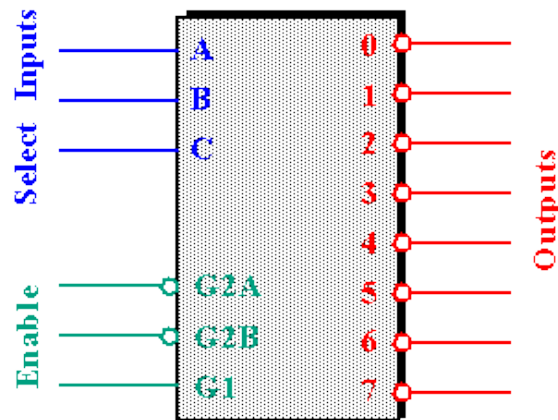


Memory Address Decoding: The 3-to-8 Line Decoder (74LS138)



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Note that all three Enables (G2A, G2B, and G1) must be active, e.g. low, low and high, respectively. Each output of the decoder can be attached to an 2764 EPROM (8K X 8).

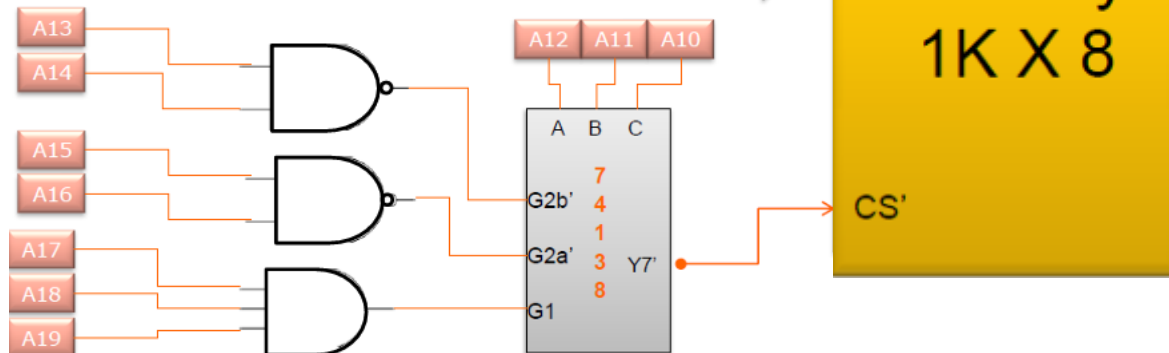


Inputs						Output							
Enable			Select										
G2A	G2B	G1	C	B	A	0	1	2	3	4	5	6	7
1	X	X	X	X	X	1	1	1	1	1	1	1	1
X	1	X	X	X	X	1	1	1	1	1	1	1	1
X	X	0	X	X	X	1	1	1	1	1	1	1	1
0	0	1	0	0	0	0	1	1	1	1	1	1	1
0	0	1	0	0	1	1	0	1	1	1	1	1	1
0	0	1	0	1	0	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	0	1	1	1	1	0	1	1	1
0	0	1	1	0	1	1	1	1	1	1	0	1	1
0	0	1	1	1	0	1	1	1	1	1	1	0	1
0	0	1	1	1	1	1	1	1	1	1	1	1	0

Address decoding using Decoder

Design a 1K RAM from FFC00 using a 74LS 138 decoder.

	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0	
RAM low	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	FFC00
RAM high	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	FFFF



Address decoding and memory interfacing

Practice Problem:

1. Design a 64K RAM and 64K ROM interface.
2. Design an address decoder for a 8K RAM from 1E000 and 8K ROM from FA000.

Memory Interfacing

Example: Interfacing 8KBytes of memory to 8088 using 2K memory chips

8K memory = 4 X 2K chips, Address lines = 11

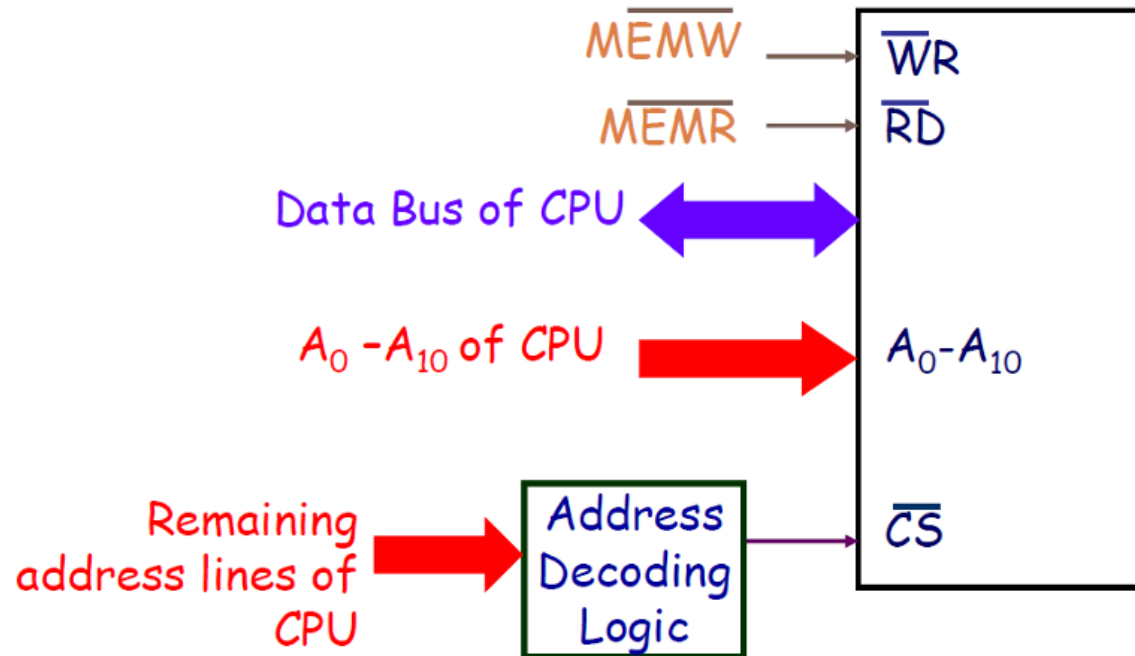
Memory Mapping : RAM1 ($00000_H - 007FF_H$) : 0000 0000 0000 0000 0000
0000 0000 0111 1111 1111

RAM2 ($00800_H - 00FFF_H$)

RAM3 ($01000_H - 017FF_H$)

RAM4 ($01800_H - 01FFF_H$)

Memory Interfacing



Memory Interfacing

RAM₁

00000_H - 007FF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1

RAM₂

00800_H - 00FFF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1

Memory Interfacing

RAM₃

01000_H-017FF_H

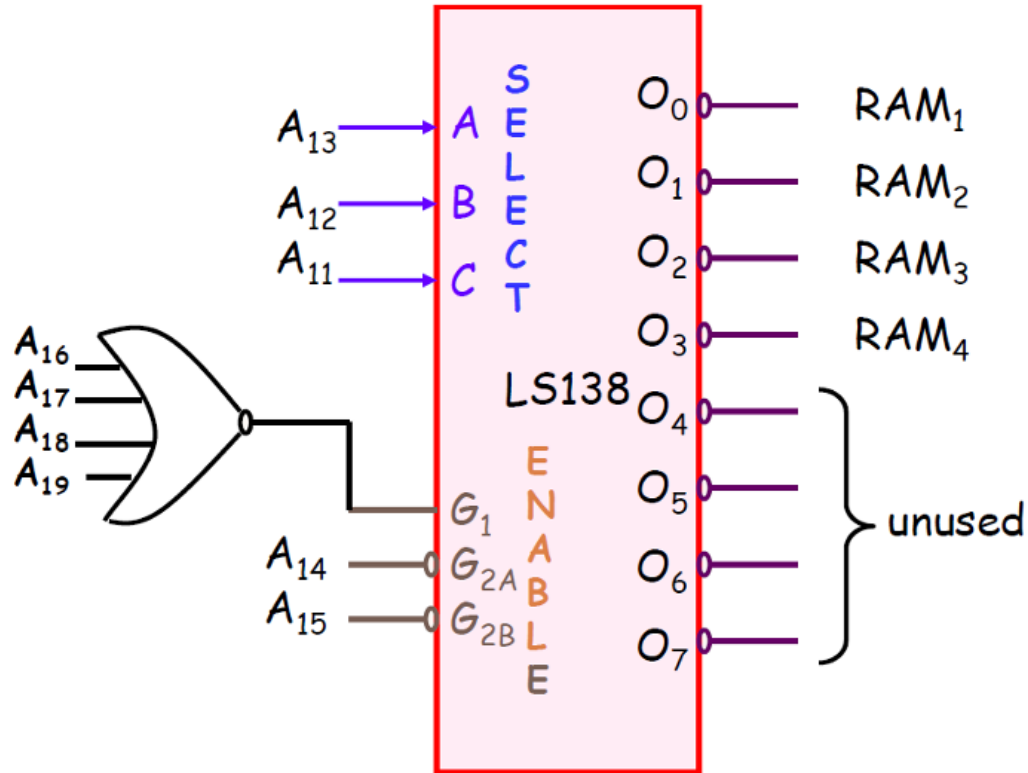
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1

RAM₄

01800_H-01FFF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1

Memory Interfacing



Memory Interfacing

Interface the memory to 8088 using the chips available

4K -2716 (ROM) starting at 00000H

8K - 6116 (SRAM) starting at 08000H

2716 ROM size 2K so Number of ROMs required -2

6116 RAM size 2K so number of RAMs required - 4

Memory Interfacing

Memory map:

ROM₁ - 00000_H - 007FF_H

ROM₂ - 00800_H - 00FFF_H

RAM₁ - 08000_H - 087FF_H

RAM₂ - 08800_H - 08FFF_H

RAM₃ - 09000_H - 097FF_H

RAM₄ - 09800_H - 09FFF_H

Memory Interfacing

ROM₁

00000_H - 007FF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1

ROM₂

00800_H - 00FFF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1

RAM₁

08000_H - 087FF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1

Memory Interfacing

RAM₂

08800_H - 08FFF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1

RAM₃

09000_H - 097FF_H

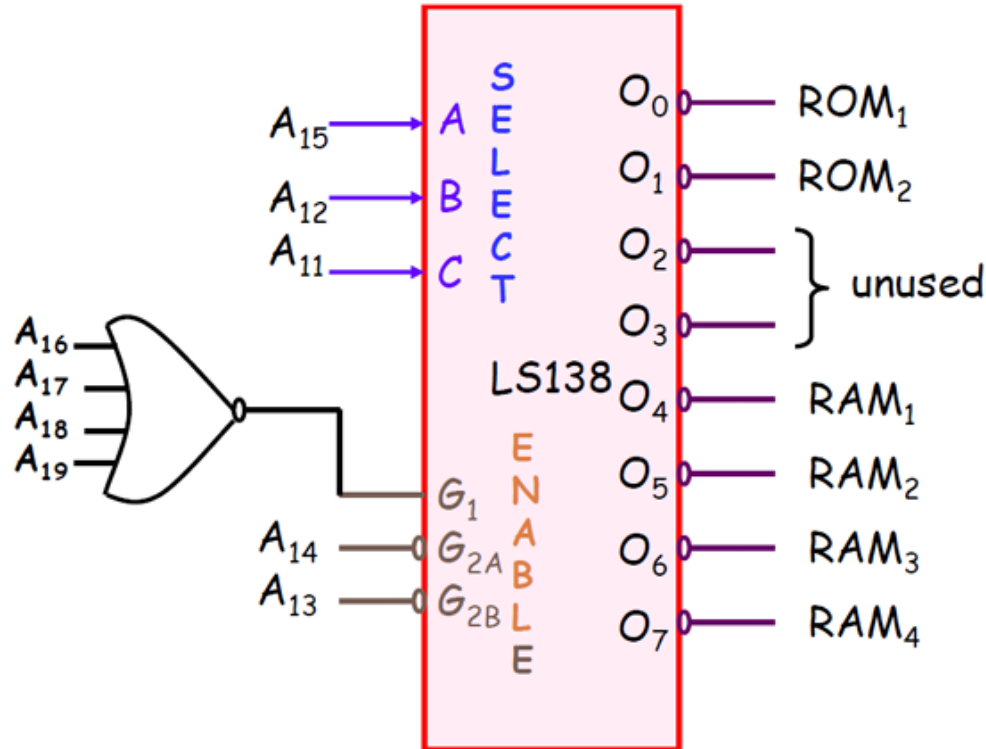
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1

RAM₄

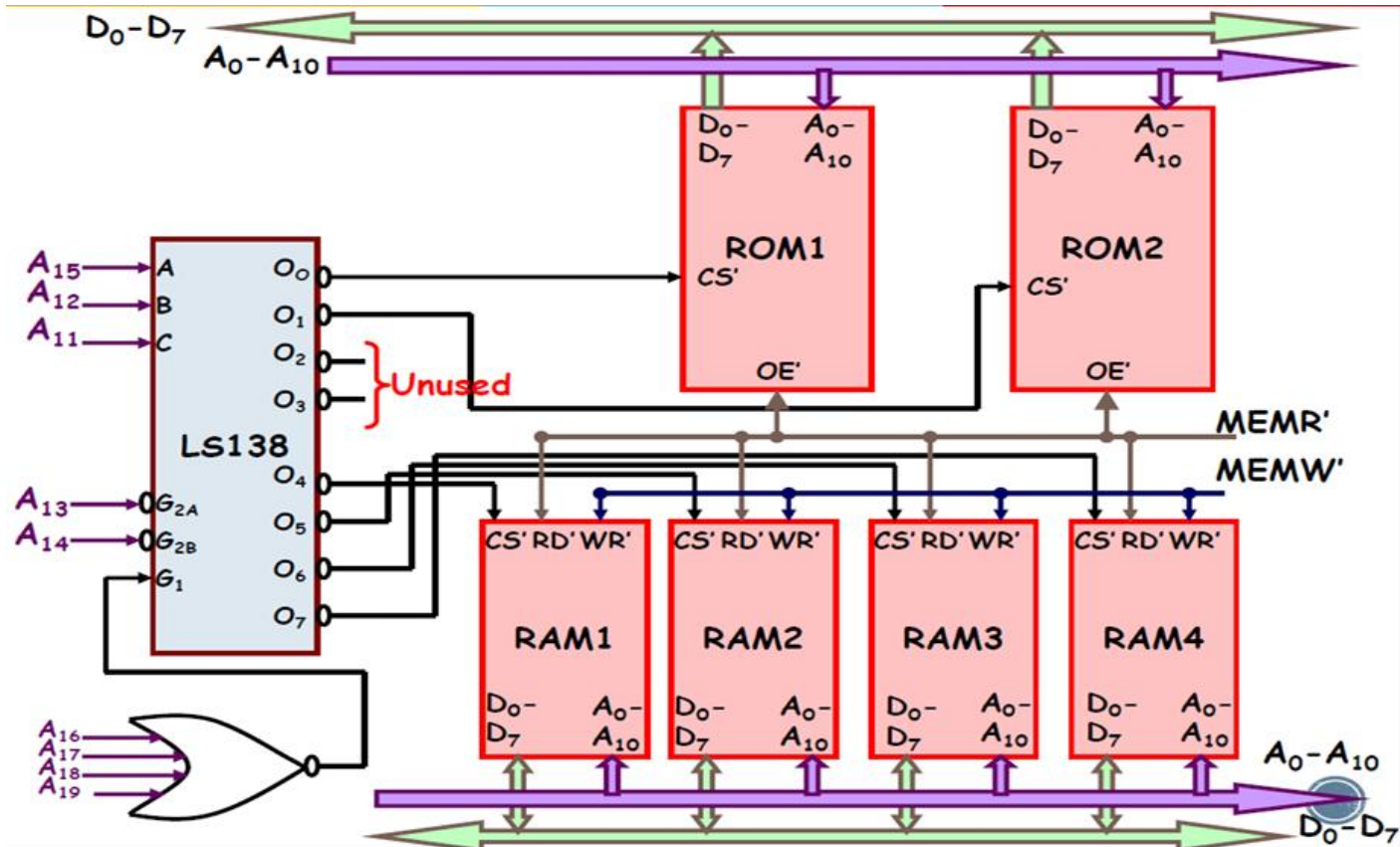
09800_H - 09FFF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1

Memory Interfacing



Memory Interfacing



Memory Interfacing

Design an interfacing circuit for 64K X 8 memory using eight 2764 EPROMs

2764 EPROM size is
8K x 8 so number of
address lines 13.

